Case Study 10: Run-Length String Compression & Efficiency Evaluator

1. Problem Statement

Develop a Python program to perform lossless Run-Length Encoding (RLE) data compression on a contiguous sequence of characters and evaluate the compression efficiency. The program prompts the user for an unspaced text string, checks for empty input, counts consecutive repeating characters to generate an encoded string of character-count pairs, displays the length before and after transformation, and determines whether the compressed representation yields a net reduction in storage size.

2. Objectives

- To validate string input against empty string submissions.
- To implement the classical Run-Length Encoding (RLE) sequential compression algorithm using string iteration and running counters.
- To dynamically append character counts and properly handle termination boundaries for the final character run.
- To compare the original and encoded lengths to evaluate and report whether data compression is beneficial.

3. Assumptions

- The input string contains no whitespace characters.
- Compression encodes repeating runs in the format <char><count> (e.g., "aaa" becomes "a3").
- Single, non-repeating characters are also appended with their frequency count of 1 (e.g., "a" becomes "a1").
- Compression is considered beneficial strictly when the length of the encoded string is less than the length of the original string (len(encoded) < len(text)).
- If len(encoded) >= len(text), the compression is categorized as not beneficial due to overhead.

4. Algorithm

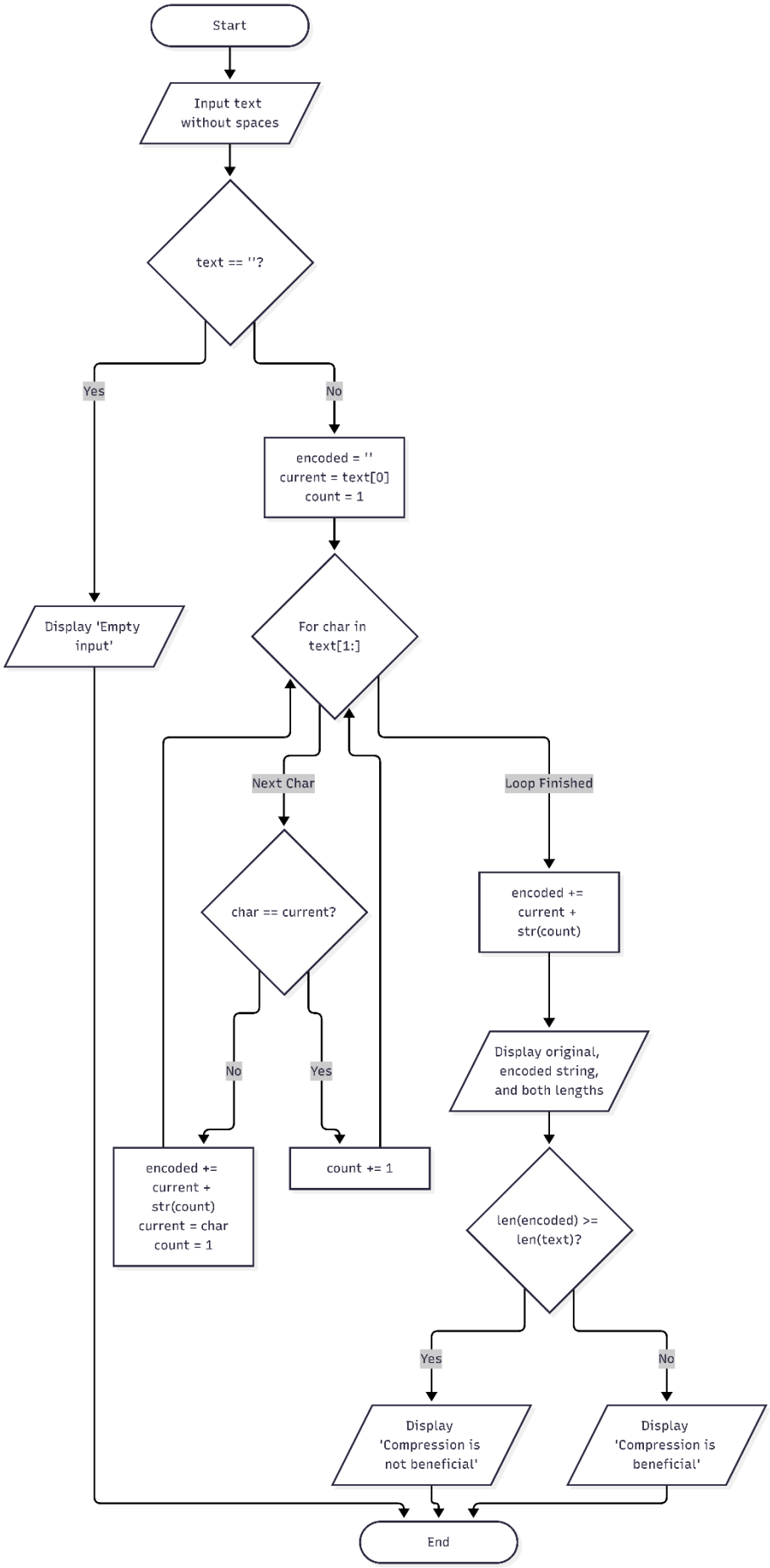
- **Step 1:** Start the program.

- **Step 2:** Prompt the user to enter a text string without spaces: text = input(...).
- **Step 3:** If text == "", print "Empty input" and proceed to Step 10.
- **Step 4:** Initialize encoded = "", set tracking variable current = text[0], and counter count = 1.
- **Step 5:** Iterate through the slice of remaining characters text[1:] using variable char:
    - If char == current: increment count += 1.
    - Else:
        - Append run to accumulator: encoded += current + str(count).
        - Reset tracking: current = char.
        - Reset counter: count = 1.
- **Step 6:** Append the final pending run to the accumulator: encoded += current + str(count).
- **Step 7:** Display the original string and the encoded string.
- **Step 8:** Display the lengths of both original (len(text)) and encoded (len(encoded)) strings.

- **Step 9:** If len(encoded) >= len(text), print "Compression is not beneficial"; otherwise, print "Compression is beneficial".

- **Step 10:** Stop the program.

5. Flowchart Logic / Diagram


Start
Input text without spaces
text == ''?
Yes
No
Display 'Empty input'
encoded = ''
current = text[0]
count = 1
For char in text[1:]
Next Char
Loop Finished
char == current?
No
Yes
encoded += current + str(count) current = char count = 1
count += 1
encoded += current + str(count)
Display original, encoded string, and both lengths
len(encoded) >= len(text)?
Yes
No
Display 'Compression is not beneficial'
Display 'Compression is beneficial'
End

6. Pseudocode

Plaintext

```
BEGIN
  PRINT "Enter text without spaces: "
  READ text

  IF text == "" THEN
    PRINT "Empty input"
  ELSE
    SET encoded = ""
    SET current = text[0]
    SET count = 1

    FOR EACH char IN text[1:] DO
      IF char == current THEN
        count = count + 1
      ELSE
        encoded = encoded + current + STRING(count)
        current = char
        count = 1
      END IF
    END FOR

    encoded = encoded + current + STRING(count)
```

```
    PRINT "Original:", text
    PRINT "Encoded:", encoded
    PRINT "Original length:", LENGTH(text)
    PRINT "Encoded length:", LENGTH(encoded)

    IF LENGTH(encoded) >= LENGTH(text) THEN
      PRINT "Compression is not beneficial"
    ELSE
      PRINT "Compression is beneficial"
    END IF
  END IF
END
```

7. Python Source Code

Python

```
# Run-Length String Compression & Efficiency Evaluator

# Read string input from user
text = input("Enter text without spaces: ")

# Check for empty string submission
if text == "":
  print("Empty input")
```

```
else:
    encoded = ""
    current = text[0]
    count = 1

    # Traverse characters starting from index 1
    for char in text[1:]:
        if char == current:
            count += 1
        else:
            encoded += current + str(count)
            current = char
            count = 1

    # Append the final sequence run
    encoded += current + str(count)

    # Display transformation metrics
    print("Original:", text)
    print("Encoded:", encoded)
    print("Original length:", len(text))
    print("Encoded length:", len(encoded))

    # Evaluate compression efficiency
    if len(encoded) >= len(text):
        print("Compression is not beneficial")
    else:
        print("Compression is beneficial")
```

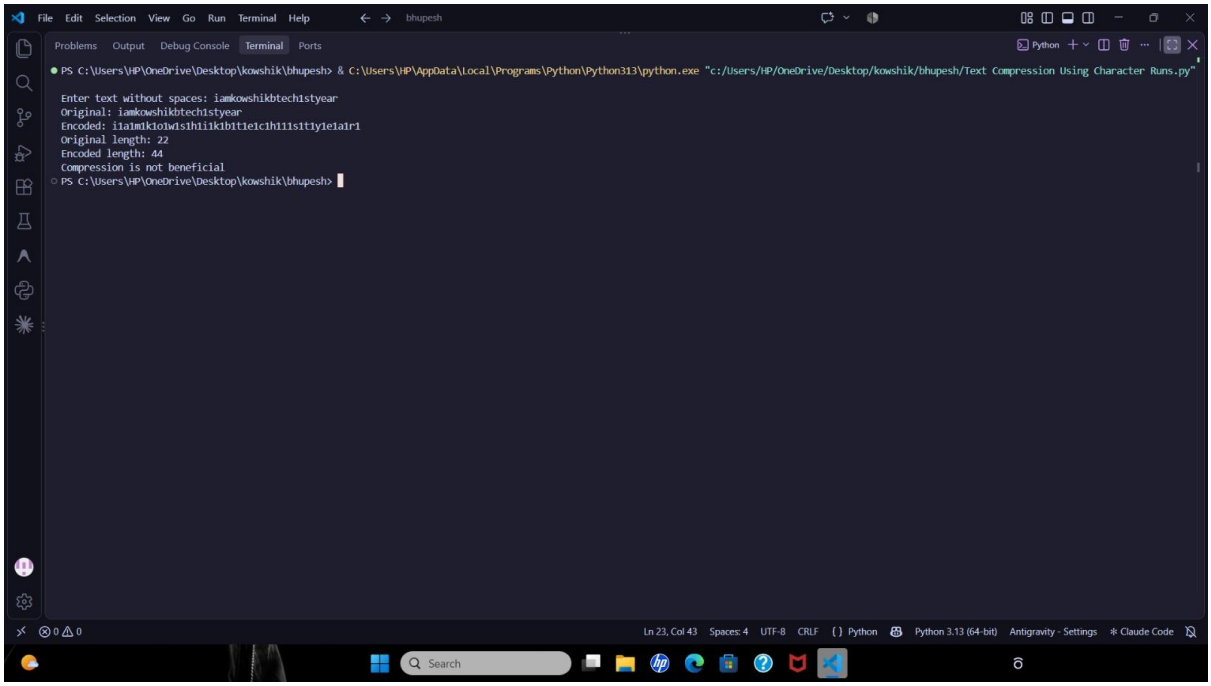
8. Test Cases

Test Case ID	Input Given	Expected Output	Actual Output	Status
TC_01 (Beneficial)	aaaaabbbbcccc	Original: aaaaabbbbcccc Encoded: a5b4c4 Original length: 13 Encoded length: 6 Compression is beneficial	Original: aaaaabbbbcccc Encoded: a5b4c4 Original length: 13 Encoded length: 6 Compression is beneficial	Pass
TC_02 (Not Beneficial)	abcdef	Original: abcdef	Original: abcdef	Pass

Test Case ID	Input Given	Expected Output	Actual Output	Status
		Encoded: a1b1c1d1e1f1 Original length: 6 Encoded length: 12 Compression is not beneficial	Encoded: a1b1c1d1e1f1 Original length: 6 Encoded length: 12 Compression is not beneficial	
TC_03 (Equal Length / Neutral)	aabb	Original: aabb Encoded: a2b2 Original length: 4 Encoded length: 4	Original: aabb Encoded: a2b2 Original length: 4 Encoded length: 4	Pass

Test Case ID	Input Given	Expected Output	Actual Output	Status
		Compression is not beneficial	Compression is not beneficial	
TC_04 (Single Character)	x	Original: x Encoded: x1 Original length: 1 Encoded length: 2 Compression is not beneficial	Original: x Encoded: x1 Original length: 1 Encoded length: 2 Compression is not beneficial	Pass
TC_05 (Empty Input)	*(Press Enter)*	Empty input	Empty input	Pass

9. Output Screenshots



10. Conclusion

The Run-Length String Compression program was executed and verified successfully. The script demonstrates the implementation of sequential character inspection using string slicing (text[1:]), run-length aggregation, state tracking, and string length comparisons to assess data compression feasibility.